

Scenario

Your organization's security team has detected a surge in suspicious network activity. There are concerns that LLMNR (Link-Local Multicast Name Resolution) and NBT-NS (NetBIOS Name Service) poisoning attacks may be occurring within your network. These attacks are known for exploiting these protocols to intercept network traffic and potentially compromise user credentials. Your task is to investigate the network logs and examine captured network traffic.

Task # 01

In the context of the incident described in the scenario, the attacker initiated their actions by taking advantage of benign network traffic from legitimate machines. Can you identify the specific mistyped query made by the machine with the IP address 192.168.232.162?

Firstly, I filtered only LLMNR traffic using filter given below.

```
udp.port == 5355
```

After filtering, mistyped queries are clearly visible and `fileshaare` is our answer.

No.	Time	Source	Destination	Protocol	Length	Info
52	2023-10-21 20:27:45.450285	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0xae2 A fileshaare
53	2023-10-21 20:27:45.450696	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0xae2 AAAA fileshaare
55	2023-10-21 20:27:45.454202	192.168.232.215	192.168.232.162	LLMNR	96	Standard query response 0xae2 A fileshaare A 192.168.232.215
56	2023-10-21 20:27:45.458616	192.168.232.215	192.168.232.162	LLMNR	108	Standard query response 0xae2 AAAA fileshaare AAAA fe80::c0a9:714f:8ea7:3313
69	2023-10-21 20:27:45.500522	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0x61e8 A fileshaare
70	2023-10-21 20:27:45.501203	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0x9b15 AAAA fileshaare
71	2023-10-21 20:27:45.503509	192.168.232.215	192.168.232.162	LLMNR	96	Standard query response 0x61e8 A fileshaare A 192.168.232.215
72	2023-10-21 20:27:45.508113	192.168.232.215	192.168.232.162	LLMNR	108	Standard query response 0x9b15 AAAA fileshaare AAAA fe80::c0a9:714f:8ea7:3313
76	2023-10-21 20:27:45.513354	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0xb281 A fileshaare
77	2023-10-21 20:27:45.513967	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0x6108 AAAA fileshaare
79	2023-10-21 20:27:45.516535	192.168.232.215	192.168.232.162	LLMNR	96	Standard query response 0xb281 A fileshaare A 192.168.232.215
80	2023-10-21 20:27:45.520834	192.168.232.215	192.168.232.162	LLMNR	108	Standard query response 0x6108 AAAA fileshaare AAAA fe80::c0a9:714f:8ea7:3313
84	2023-10-21 20:27:45.532216	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0xc4a3 A fileshaare
85	2023-10-21 20:27:45.532733	192.168.232.162	224.0.0.252	LLMNR	70	Standard query 0x1ce7 AAAA fileshaare
87	2023-10-21 20:27:45.535612	192.168.232.215	192.168.232.162	LLMNR	96	Standard query response 0xc4a3 A fileshaare A 192.168.232.215
88	2023-10-21 20:27:45.540065	192.168.232.215	192.168.232.162	LLMNR	108	Standard query response 0x1ce7 AAAA fileshaare AAAA fe80::c0a9:714f:8ea7:3313
168	2023-10-21 20:30:45.273518	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x4a65 A printe
169	2023-10-21 20:30:45.273953	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x5ae5 AAAA printe

Answer: `fileshaare`

Task # 02

We are investigating a network security incident. To conduct a thorough investigation, We need to determine the IP address of the rogue machine. What is the IP address of the machine acting as the rogue entity?

we can clearly see in the filtered traffic that when mistyped query is sent (fileshaare). The IP responding to these queries is attacker's IP.

No.	Time	Source	Destination	Protocol	Length	Info
88	2023-10-21 20:27:45.540065	192.168.232.215	192.168.232.162	LLMNR	108	Standard query response 0x1ce7 AAAA fileshaare AAAA fe80::c0a9:714f:8ea7:3313
168	2023-10-21 20:30:45.273518	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x4a65 A printr
169	2023-10-21 20:30:45.273953	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x5ae5 AAAA printr
171	2023-10-21 20:30:45.276421	192.168.232.215	192.168.232.176	LLMNR	90	Standard query response 0x4a65 A printr A 192.168.232.215
172	2023-10-21 20:30:45.280843	192.168.232.215	192.168.232.176	LLMNR	102	Standard query response 0x5ae5 AAAA printr AAAA fe80::c0a9:714f:8ea7:3313
185	2023-10-21 20:30:45.324555	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x3188 A printr
186	2023-10-21 20:30:45.325032	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x567d AAAA printr
187	2023-10-21 20:30:45.327277	192.168.232.215	192.168.232.176	LLMNR	90	Standard query response 0x3188 A printr A 192.168.232.215
188	2023-10-21 20:30:45.331321	192.168.232.215	192.168.232.176	LLMNR	102	Standard query response 0x567d AAAA printr AAAA fe80::c0a9:714f:8ea7:3313
192	2023-10-21 20:30:45.338478	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x02c2 A printr
194	2023-10-21 20:30:45.339381	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0xb232 AAAA printr
195	2023-10-21 20:30:45.341756	192.168.232.215	192.168.232.176	LLMNR	90	Standard query response 0x02c2 A printr A 192.168.232.215
196	2023-10-21 20:30:45.345988	192.168.232.215	192.168.232.176	LLMNR	102	Standard query response 0xb232 AAAA printr AAAA fe80::c0a9:714f:8ea7:3313
200	2023-10-21 20:30:45.351708	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0x85e5 A printr
201	2023-10-21 20:30:45.352077	192.168.232.176	224.0.0.252	LLMNR	67	Standard query 0xd22d AAAA printr
203	2023-10-21 20:30:45.354254	192.168.232.215	192.168.232.176	LLMNR	90	Standard query response 0x85e5 A printr A 192.168.232.215
204	2023-10-21 20:30:45.357064	192.168.232.215	192.168.232.176	LLMNR	102	Standard query response 0xd22d AAAA printr AAAA fe80::c0a9:714f:8ea7:3313

Answer: 192.168.232.215

Task # 03

As part of our investigation, identifying all affected machines is essential. What is the IP address of the second machine that received poisoned responses from the rogue machine?

The IP mentioned in the Task 1 is one machine which is affected, Other affected machine is the one making queries for printr .

Answer: 192.168.232.176

Task # 04

We suspect that user accounts may have been compromised. To assess this, we must determine the username associated with the compromised account. What is the username of the account that the attacker compromised?

For this task, we need to filter traffic for ntlmssp . This protocol is used for authenticatin in windows environment, so we can identify username of the account by filtered traffic.

No.	Time	Source	Destination	Protocol	Length	Info
240	2023-10-21 20:33:09.559591	192.168.232.215	192.168.232.176	SMB2	240	Session Setup Request, NTLMSSP NEGOTIATE
241	2023-10-21 20:33:09.560775	192.168.232.176	192.168.232.215	SMB2	453	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP CHALLENGE
242	2023-10-21 20:33:09.570612	192.168.232.215	192.168.232.176	SMB2	598	Session Setup Request, NTLMSSP AUTH, User: cybercactus.local\janesmith

Answer: janesmith

Task # 05

As part of our investigation, we aim to understand the extent of the attacker's activities. What is the hostname of the machine that the attacker accessed via SMB?

For this task, we need to filter traffic for smb2 protocol as it is stated that smb activities happened after compromise. Looking at the traffic, we can notice the session being created between machines. By expanding NTLM properties of the packets involving session creation, we can clearly get the hostname of the machine.

The image shows a Wireshark capture of SMB2 traffic. The packet list on the left shows a session setup request (packet 241) and a session setup response (packet 242). The packet details pane for packet 242 shows the NTLM Server Challenge and Target Info. The Target Info section includes the NetBIOS domain name (CYBERCACTUS), NetBIOS computer name (ACCOUNTINGPC), DNS domain name (cybercactus.local), and DNS tree name (AccountingPC.cybercactus.local). The packet bytes pane on the right shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
223	2023-10-21 20:32:08.280679	192.168.232.176	192.168.232.148	SMB2	178	Ioctl Request FSCTL_QUERY_NETWORK_INTERFACE_INFO
224	2023-10-21 20:32:08.280920	192.168.232.148	192.168.232.176	SMB2	474	Ioctl Response FSCTL_QUERY_NETWORK_INTERFACE_INFO
227	2023-10-21 20:32:18.755248	192.168.232.176	192.168.232.148	SMB2	126	Tree Disconnect Request
228	2023-10-21 20:32:18.755522	192.168.232.148	192.168.232.176	SMB2	126	Tree Disconnect Response
229	2023-10-21 20:32:18.755786	192.168.232.176	192.168.232.148	SMB2	126	Session Logoff Request
230	2023-10-21 20:32:18.755892	192.168.232.148	192.168.232.176	SMB2	126	Session Logoff Response
236	2023-10-21 20:33:09.525590	192.168.232.176	192.168.232.215	SMB2	306	Negotiate Protocol Response
238	2023-10-21 20:33:09.545042	192.168.232.215	192.168.232.176	SMB2	266	Negotiate Protocol Request
239	2023-10-21 20:33:09.545778	192.168.232.176	192.168.232.215	SMB2	308	Negotiate Protocol Response
240	2023-10-21 20:33:09.559591	192.168.232.215	192.168.232.176	SMB2	240	Session Setup Request, NTLMSSP NEGOTIATE
241	2023-10-21 20:33:09.560775	192.168.232.176	192.168.232.215	SMB2	453	Session Setup Response, Error: STATUS_MORE_PROCESSING_REQUIRED, NTLMSSP CHALLENGE
242	2023-10-21 20:33:09.570612	192.168.232.215	192.168.232.176	SMB2	598	Session Setup Request, NTLMSSP_AUTH, User: cybercactus.local\janesmith
250	2023-10-21 20:33:09.685817	192.168.232.176	192.168.232.215	SMB2	139	Session Setup Response
251	2023-10-21 20:33:09.698065	192.168.232.215	192.168.232.176	SMB2	230	Encrypted SMB3
252	2023-10-21 20:33:09.699505	192.168.232.176	192.168.232.215	SMB2	190	Encrypted SMB3
257	2023-10-21 20:33:09.712835	192.168.232.215	192.168.232.176	SMB2	178	Encrypted SMB3
258	2023-10-21 20:33:09.713191	192.168.232.176	192.168.232.215	SMB2	178	Encrypted SMB3

NTLM Server Challenge: c8eb5e4b7c037617
Reserved: 0000000000000000

Target Info

- Length: 210
- Maxlen: 210
- Offset: 78
- Attribute: NetBIOS domain name: CYBERCACTUS
- Attribute: NetBIOS computer name: ACCOUNTINGPC
- Attribute: DNS domain name: cybercactus.local
- Attribute: DNS computer name: AccountingPC.cybercactus.local
- Attribute: DNS tree name: cybercactus.local
- Attribute: Timestamp
- Attribute: End of List

Version 10.0 (Build 19041); NTLM Current Revision 15
Major Version: 10
Minor Version: 0
Build Number: 19041
NTLM Current Revision: 15

SMB2 (Server Message Block Protocol version 2): Protocol

Packets: 269 · Displayed: 41 (15.2%)

Profile: Default

Answer: AccountingPC